

Linear Optimization

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CS2S MTECH12009



Linear Optimizatⁿ

24th Sept, 2025

Primal Problem

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \begin{bmatrix} C_1 & C_2 & \dots & C_n \\ \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} A \\ \end{bmatrix} \begin{bmatrix} C \\ \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} B \end{bmatrix}$$
 Every column becomes a constraint

Two:

$$\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & & & 0 \\ 0 & & & & \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

To
define
Qual

Fit's Qual

$$a_{11}y_1 + a_{21}y_2 + \dots + a_{k+1,1}y_{k+1} + a_{m1}y_m \geq c_1$$

$$\Rightarrow a_{12}y_1 + a_{22}y_2 + \dots + a_{m2}y_m \geq c_2$$

$$a_{1n}y_1 + a_{2n}y_2 + \dots + a_{mn}y_m \geq C_n$$

$$\begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & & & & 1 \end{bmatrix} \begin{matrix} \geq 0 \\ \geq 0 \\ \geq 0 \end{matrix}$$

→ We'll get same objective

K tight Rows

All optimum point

$$\begin{matrix} y_1 = \alpha_1 \\ y_2 = \alpha_2 \\ y_k = 0 \\ \vdots \\ y_m = 0 \end{matrix} \begin{bmatrix} a_{11}x_1 & a_{12}x_2 & \dots & a_{1n}x_n \\ a_{21}x_1 & a_{22}x_2 & \dots & a_{2n}x_n \\ a_{k1}x_1 + a_{k2}x_2 & & & a_{kn}x_n \\ a_{k+11}x_1 + a_{k+12}x_2 & & & a_{k+1n}x_n \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \leq \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \quad (x_1, \dots, x_n)$$

$$\begin{bmatrix} -1x_1 & 0 & \dots & 0 \\ 0 & -1x_2 & & \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & & -1x_n \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}$$

multiply by y_1

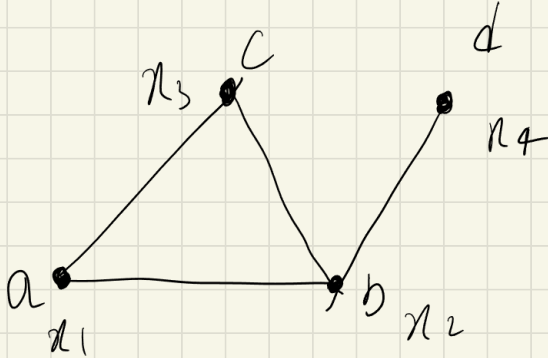
$$\begin{bmatrix} a_{11}x_1 & a_{12}x_2 & \dots & a_{1n}x_n \\ a_{21}x_1 & a_{22}x_2 & \dots & a_{2n}x_n \\ \dots & \dots & \dots & \dots \\ a_{k1}x_1 + a_{k2}x_2 & \dots & \dots & a_{kn}x_n \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \leq \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}$$

$$\begin{array}{c} \text{C}_1 x_1 \quad \text{C}_2 x_2 \quad \dots \quad \text{C}_n x_n \end{array} \begin{bmatrix} a_{11}x_1y_1 + a_{12}x_2y_1 + \dots + a_{1n}x_1y_1 \\ \vdots \\ a_{k1}x_1y_k + a_{k2}x_2y_k + \dots + a_{kn}x_1y_k \\ \dots \\ a_{k+1,1}x_1y_{k+1}^{(0)} + \dots + a_{k+1,n}x_{k+1}^{(0)}y_{k+1} \\ \vdots \\ 0 \quad 0 \quad 0 \quad 0 \end{bmatrix} \begin{bmatrix} by_1 \\ \vdots \\ by_k \\ \vdots \\ b_{k+1}^{(0)} \\ \vdots \\ b_{m \times 0} \end{bmatrix}$$

Algebra completed, now coming
Back to CSE.

Vertex Cover Problem

We changed from Graph to $\mathbb{R}^0$



$$\begin{aligned} x_1 + x_2 &\geq 1 \\ x_2 + x_3 &\geq 1 \end{aligned}$$

$$x_1 + x_3 \geq 1$$

$$x_2 + x_4 \geq 1$$

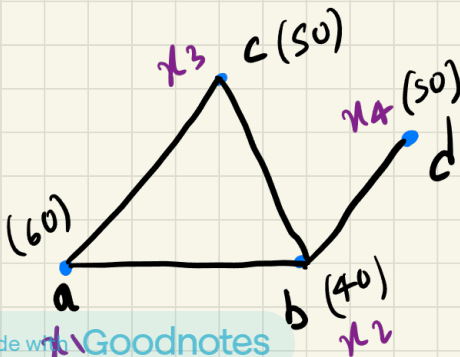
$$x_i \in \{0, 1\}$$

Minimizingⁿ of $x_1 + x_2 + x_3 + x_4$
objective $\uparrow$

$\rightarrow$ We relaxed this condⁿ $0 \leq x_i \leq 1$

25th Sept, 2025

$$x_i \in \{0, 1\}$$



$$\begin{aligned} x_1 + x_2 &\geq 1 \\ x_1 + x_3 &\geq 1 \\ x_2 + x_3 &\geq 1 \\ x_2 + x_4 &\geq 1 \end{aligned}$$

Minimize $(60x_1 + 40x_2 + 50x_3 + 50x_4)$

$x_i \in \{0, 1\}$

$x_i \geq 0 \rightarrow$ LP constraint

→ Change Primal → Dual

→ Solve Dual

→ W.K.T Primal Solⁿ = Dual Solⁿ

Dual

Minimize $(60x_1 + 40x_2 + 50x_3 + 50x_4)$

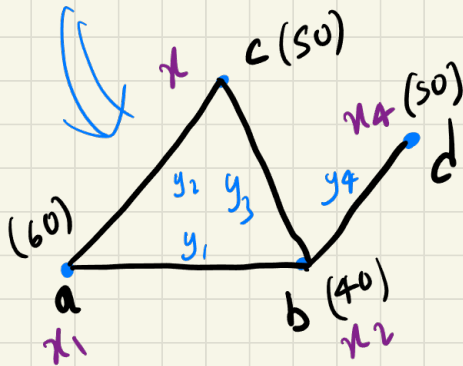
y_1	x_1	+	x_2			≥ 1
y_2	x_1	+		x_3		≥ 1
y_3			x_2	+	x_3	≥ 1
y_4			x_2	+	x_4	≥ 1

$y_i \geq 0$ Objective

$\max(y_1 + y_2 + y_3 + y_4)$

Constraints

We're giving weights to graph.



$y_1 + y_2 \leq 60$

$y_1 + y_3 + y_4 \leq 40$

$y_2 + y_3 \leq 50$

$y_4 \leq 50$

We can solve Dual

After solving, see if the sum of edges == vertex cost
Include that

In short, search equality here

$y_1 + y_2 \leq 60$

$y_1 + y_3 + y_4 \leq 40$

$y_2 + y_3 \leq 50$

$y_4 \leq 50$

Vertex Cover: {

If $y_1 + y_2 = 60$ } a

If $y_1 + y_3 = 40$ } b

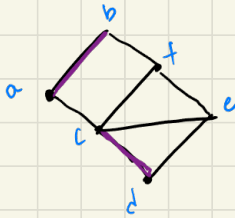
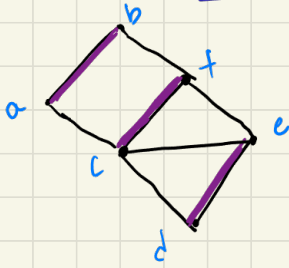
If $y_4 = 50$ } d

If $y_2 + y_3 = 50$ } c

6th October, 2025

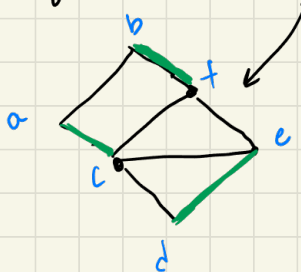
Matching :- set of edges such that no common vertex.

$cf, de \rightarrow$ Matching set
 $ab, cf, de \rightarrow$ Max Matching Graph can have many mm.



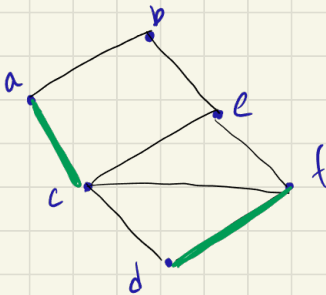
Alternate Path $\rightarrow$ ' $\overline{b} \underline{a} \overline{c} \underline{d}$ ' Not part part of matching

Augmenting path $\rightarrow$ $\textcircled{f} \underline{b} \underline{a} \underline{c} \underline{d} \textcircled{e}$ Not part of any matching



size is increased by 1.

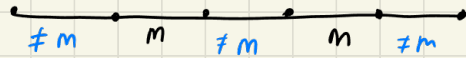
(dunder 18^m & 19^m Lec)



$G(V, E)$

MCE (M is not mm)

such path exists $\rightarrow$ (augmenting path)



Increase the size by exactly one.

$M \quad |M'| > |M|$

$G' = (V, (M' - M) \cup (M - M'))$ symmetric diff of M' & M [$M' \oplus M$]

$\max \text{ degree } (G') \leq 2$

3 Cases

G' can

be decompos.

into



Cycles of even length



Isolated vertices



Path $\rightarrow$ P_e (Even length path)

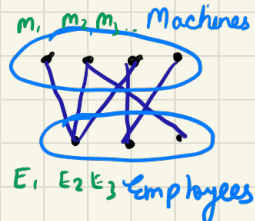
P_o (Odd length path)



$\rightarrow$ This is augmenting path

$\rightarrow$ Keep doing this, untill you can't add, you'll get maximum Matching

9th October, 2025



Input : Graph G (We'll take bipartite graph)

Objective : Maximise the assignment

We'll start by ϕ , will use augmenting path, extend by 1.

- Repeat process
- Modify BFS Algorithm slightly

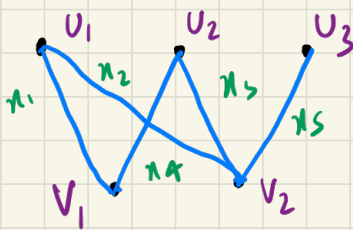
How?

Home work

$A \leftarrow G(V, E)$ → Bipartite graph
 $A \leftarrow h(V, E), M$ should give augmenting path (Algorithm is slightly tampering of BFS)

m_0, ϕ
 m_1, m_2
 m_2, m_3 max match
 $\vdots$
 $m_{n-1}, 0$

Input → Bipartite graph : (Can you express this as ILP?)



- Constraint?
- Variable?
- Objective f^* ?

Integer Linear Program

- Assign $\{0, 1\}$ for every edge (Part of matching or not)
- $x_i \in \{0, 1\}$
- Maximise → $\sum x_i$ Relax it $0 \leq x_i \leq 1$ for our simplex algorithm

→ One person → One machine (And vice versa)

$$x_1 + x_4 \leq 1$$

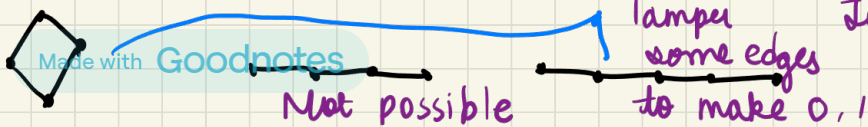
$$x_1 + x_4 + x_5 \leq 1$$

$$x_1 + x_2 \leq 1$$

$$x_5 \leq 1$$

$$x_4 + x_3 \leq 1$$

- How to get our solⁿ $x_i \in \{0, 1\} \Leftrightarrow 0 \leq x_i \leq 1$ (How?)
- Fix 0 or 1 (if 0 OR 1) → We can have cycles if it'll be even length

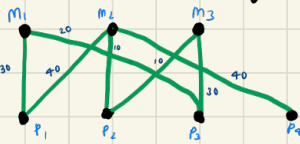


10th October, 2025

ILP $\hat{=}$ Relaxed version (same solⁿ)

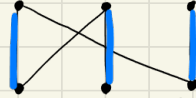
⊙ Cycle $\rightarrow$ Tamper edge weight (wt. becomes 0 & 1)

⊙ $\rightarrow$ Odd length path not possible $\rightarrow$ weight is maximised



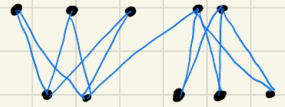
Now, maximise the profit (not assignments)
Formulate ILP $\Rightarrow$ Solve $\Rightarrow$ Relax

Perfect Matching -

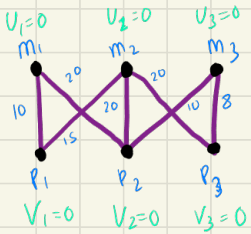


Perfect Matching

$\Rightarrow$ All graph doesn't have perfect matching.



not possible



Edge weights are my expenses; reduce it.

$\rightarrow$ Find a perfect matching with least cost.

Primal $\rightarrow$ Minimize (Edge)

Dual $\rightarrow$ Maximize (Vertex)

$\rightarrow$ Make another smaller graph, with tight edges

Constraints

$$U_1 + V_1 \leq 10$$

$$V_1 + U_2 \leq 15$$

$$V_2 + U_2 \leq 20$$

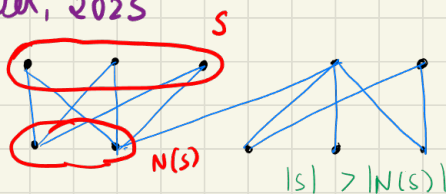
$\vdots$

Assign weights to the vertex.



$\rightarrow$ see the augmenting part proof!!

15th October, 2025



For a graph G , $\exists S \subset V$
 $|S| > |N(S)|$

If I can find something S .
 we can't find perfect matching here.

→ same augmenting path idea is used here.

Dual cost

$$\max (u_1 + u_2 + u_3 + u_4 + v_1 + v_2 + v_3 + v_4)$$

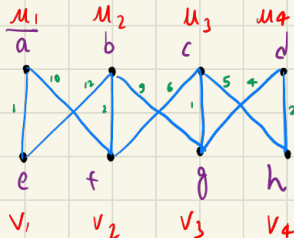
⇒ Condⁿ Sum of vertex $\leq$ edge wt.

Dual $\leq$ Primal (Proof 2)

Primal cost

min (Edge weight)

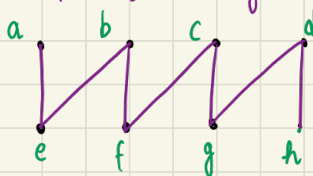
Every level, it'll increase by one.



$$\begin{cases} u_1 + v_1 \leq 1 \\ u_1 + v_2 \leq 10 \\ \vdots \end{cases}$$

Dual

$u_1 + v_2 = 10$ Tight
 $u_1 + v_2 < 10$ Untight



This will have a perfect matching.

→ Bipartite Graph edge wt.

→ Find min matching

→ Dual (Edge to vertex) ; condⁿ

→ Dual $\leq$ Primal solⁿ

→ Dual = Primal

→ For a subgraph, if G_H has not perfect matching, the dual is not optimal.

$G, m \quad \exists S \subset V \quad |S| > |N(S)|$

Thursday ; Friday Off

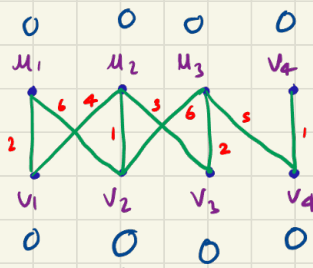


Primal $\hat{=}$ Min
 Dual $\hat{=}$ Maximizing

22nd Oct, 2025 (Wed)

4th Assignment

Compute
⇒ Max Matching
== Perfect Matching
(4) in this case



(Initial Feasible point)

Start with 0 weight

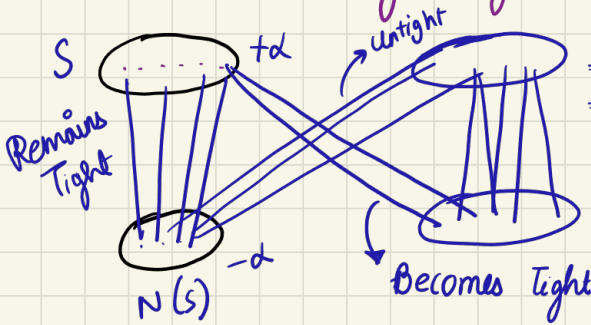
⇒ Dual lower primal bounds

↳ Tamper weights, make more tight edges.
move one solⁿ to another solⁿ
(objective ↑↑)

$|S| > |N(S)| \leftarrow$ Use this to tamper weight, so no. of tight edges

$+\alpha$ $-\alpha$ will increase

↳ Previously untight edges will become tight.



⇒ size of max. Matching increase
⇒ sum of weight also increases

Input ⇒ Bipartite

Output ⇒ Max Matching

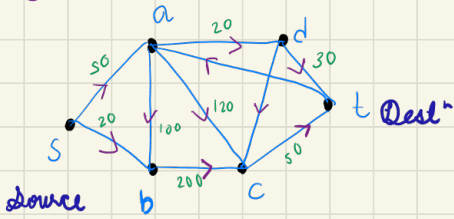
Augmenting path, max path

- Assign weight (0,0)
- Construct smaller graph
- Comput. Max Matching
- See if perf. matching
- $|S| > |N(S)|$
 $+\alpha$ $-\alpha$
- Some edge become tight (Iterate)

23rd Oct, 2025 (Thu) 10:00 AM

or store

⇒ Can't create/destroy; Just pass it
 Max B/w from source to dest?
 • → Switches
 — → Bandwidth

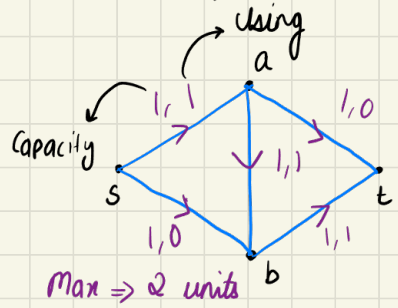


⇒ Assign variables to edges
 ⇒ Edge Vertex (Incoming == Outgoing)

Objective F^* ⇒

$$S(Out - In) =$$

$$T(In - Out)$$



Initially, assign 0 to every edge.

Can move in opposite direcⁿ as well (Decreasing)

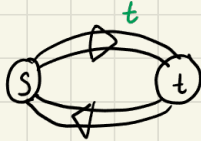
29th Oct, 2025 (11:00 AM)
Wednesday

(min bound)

Max flow = Min cut (Upper bound)

primal

Dual

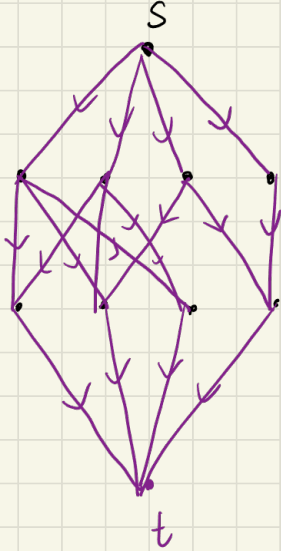


(Forward edge)

Max flow can't be more than t .

→ When algorithm terminates (previous one),

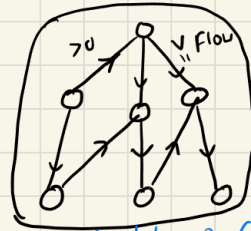
→ Max flow algo $\equiv$ Maximum Matching in Bi-partite Graph



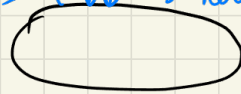
capacity = 1

capacity = ∞

capacity = 1



= 0 if $\downarrow \downarrow$ } can't go from here to there



Max Flow = Max matching in Bipartite Graph

How do you tamper the values?

→ Find cycles, tamper weights

→ End up in 0 or 1.

Not meeting tomorrow

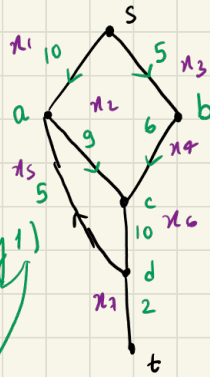
3rd Nov, 2025
(Monday)

Shortest Path from
s to t ?

→ $S = \#I - \#O = -1$

→ $T = \#I - \#O = 1$

(Exactly 1)



→ Assign $\{0,1\}$ to edge
(Include/Not included)

$x_i \in \{0,1\}$

Objective F^* :

$\min (10x_1 + 9x_2 + 5x_3 + 6x_4 + 5x_5 + 10x_6 + 2x_7)$

→ How to make sure that selected edges
form a path from source to sink.

Slight modification of n/w flow problem

$S \Rightarrow -x_1 - x_3 = -1$

$C \Rightarrow x_2 + x_4 - x_6 = 0$

$a \Rightarrow x_1 + x_5 - x_2 = 0$

$T \Rightarrow x_7 = 1$

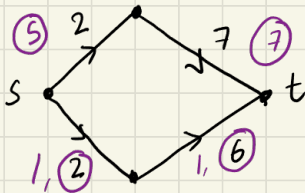
→ Constraint will make sure that
we'll get a path

Relax constraints $\begin{cases} 0 \leq x_i \leq 1 \\ x_i \in \{0,1\} \end{cases}$

Both will have same cost

→ How to write dual? & Algorithm

5th Assignment → N/w Flow (Last Class)



Min cost max flow using simplex algo.

Input → Directed graph
Cost; Flow

Output →

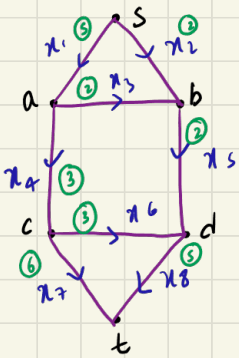
NO TESTCASES FOR $P1, P2, P3$

→ 5 ASSIGNMENT

→ 1, 2, 3 Single

→ 4, 5 Double

6th Nov, 2025 (Thursday)



Relax $0 \leq x_i \leq 1$ Tampor, don't change objective f^{*}; make 0 & 1.

$$\min (5x_1 + 2x_2 + 2x_3 + 3x_4 + 6x_5 + 3x_6 + 6x_7 + 5x_8)$$

$$x_1 - x_3 - x_4 = 0$$

$$x_1 + x_3 - x_5 = 0$$

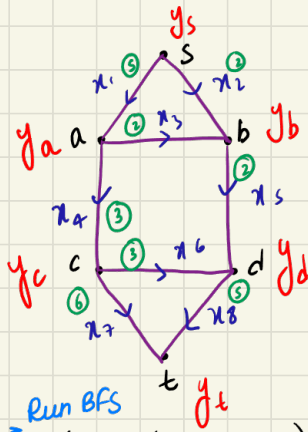
$$x_7 + x_8 = 1 \Rightarrow \text{Source}$$

$$-x_1 - x_2 = -1 \Rightarrow \text{Res^ul}$$

Qual

When constraints are equality; dual can be positive or negative.

constraint $\Rightarrow$ Variable
Variable $\Rightarrow$ Constraint



Primal $\downarrow \downarrow$
Dual $\uparrow \uparrow$

$$\max \rightarrow (y_t - y_s)$$

Constraints $\rightarrow$

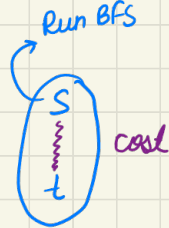
$$y_a - y_s \leq 5$$

$$y_b - y_s \leq 2$$

$$y_b - y_a \leq 2$$

$$y_c - y_a \leq 3$$

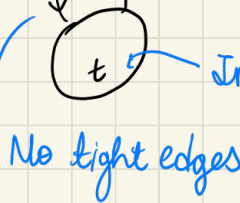
Assign a real no. to each vertex



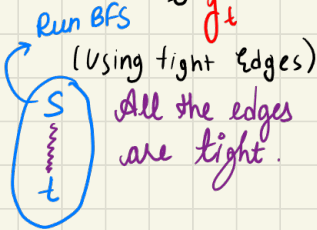
Will get a path
That is a shortest path.

$$\text{Dual} \leq \text{Primal Cost}$$

Start from edge weight 0.



Increase the weight by same amount
(Increasing tight edges)



== Course Ends here ==

